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VITMEE 2024 Question Paper

Vellore Institute of Technology Master's Entrance
Examination (VITMEE)

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- To the lines $ax^2 + 2hxy + by^2 = 0$, the lines $a^2x^2 + 2h(a+b)xy + b^2y^2 = 0$ are
 - equally inclined
 - perpendicular
 - bisector of the angle
 - None of the above
- If R be a relation from $A = \{1, 2, 3, 4\}$ to $B = \{1, 3, 5\}$ such that $(a, b) \in R \Leftrightarrow a < b$, then $R \circ R^{-1}$ is
 - $\{(1, 3), (1, 5), (2, 3), (2, 5), (3, 5), (4, 5)\}$
 - $\{(3, 1), (5, 1), (3, 2), (5, 2), (5, 3), (5, 4)\}$
 - $\{(3, 3), (3, 5), (5, 3), (5, 5)\}$
 - $\{(3, 3), (3, 4), (4, 5)\}$
- If $x + iy = (1 - i\sqrt{3})^{100}$, then find (x, y) .
 - $(2^{99}, 2^{99}\sqrt{3})$
 - $(2^{99}, -2^{99}\sqrt{3})$
 - $(-2^{99}, 2^{99}\sqrt{3})$
 - None of these
- For a GP, $a_n = 3(2^n)$, $\forall n \in N$. Find the common ratio.
 - 2
 - $1/2$
 - 3
 - $1/3$
- If a, b, c are in HP, then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ will be in
 - AP
 - GP
 - HP
 - None of these
- If $\frac{x^2 + 2x + 7}{2x + 3} < 6$, $x \in R$, then
 - $x > 11$ or $x < -\frac{3}{2}$
 - $x > 11$ or $x < -1$
 - $-\frac{3}{2} < x < -1$
 - $-1 < x < 11$ or $x < -\frac{3}{2}$
- The number of ways of painting the faces of a cube of six different colours is
 - 1
 - 6
 - 6!
 - 36
- A line passes through $(2, 2)$ and is perpendicular to the line $3x + y = 3$. What is its y -intercept?
 - $1/3$
 - $2/3$
 - 1
 - $4/3$
- The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is
 - 0
 - 1
 - 3
 - 4
- If D is the set of all real x such that $1 - e^{(1/x) - 1}$ is positive, then D is equal to
 - $(-\infty, 1]$
 - $(-\infty, 0)$
 - $(1, \infty)$
 - $(-\infty, 0) \cup (1, \infty)$
- Find the value of the limit $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos x}}{x}$.
 - 0
 - 1
 - $\sqrt{2}$
 - does not exist
- Evaluate $\int \frac{x^2 + 4}{x^4 + 16} dx$.
 - $\frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 4}{2x\sqrt{2}} \right) + C$
 - $\frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 4}{2\sqrt{2}} \right) + C$
 - $\frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 4}{x\sqrt{2}} \right) + C$
 - None of the above
- Evaluate $\int_{\pi/4}^{3\pi/4} \frac{1}{1 + \cos x} dx$
 - 2
 - 2
 - $1/2$
 - $-1/2$
- If one AM 'A' and two GM p and q are inserted between two given numbers, then find the value of $\frac{p^2}{q} + \frac{q^2}{p}$.
 - A
 - 2A
 - 3A
 - 4A
- If the roots of the equation $x^2 + ax + b = 0$ are c and d , then one of the roots of the equation $x^2 + (2c + a)x + c^2 + ac + b = 0$ is
 - c
 - $d - c$
 - $2d$
 - $2c$
- The sum of the coefficients of $(6a - 5b)^n$, where n is a positive integer, is
 - 1
 - 1
 - 2^n
 - 2^{n-1}

17. Find the value of $(7.995)^{1/3}$ correct to four decimal places.
 (a) 1.9995 (b) 1.9996
 (c) 1.9990 (d) 1.9991
18. The values of constants a and b so that $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0$ are
 (a) $a = 0, b = 0$
 (b) $a = 1, b = -1$
 (c) $a = -1, b = 1$
 (d) $a = 2, b = -1$
19. The projection of the vector $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ on the vector $4\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}$ is
 (a) $\frac{5\sqrt{6}}{10}$ (b) $\frac{19}{9}$
 (c) $\frac{9}{19}$ (d) $\frac{\sqrt{6}}{19}$
20. If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three non-zero vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$ and $m = \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}$, then
 (a) $m < 0$ (b) $m > 0$
 (c) $m = 0$ (d) $m = 3$
21. A line making angles 45° and 60° with the positive directions of the axes of x and y makes with the positive direction of z -axis, an angle of
 (a) 60° (b) 120°
 (c) 60° or 120° (d) None of these
22. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then B is equal to
 (a) $I \cos \theta + J \sin \theta$
 (b) $I \sin \theta + J \cos \theta$
 (c) $I \cos \theta - J \sin \theta$
 (d) $-I \cos \theta + J \sin \theta$
23. Which of the following is correct?
 (a) Determinant is a square matrix
 (b) Determinant is a number associated to a matrix
 (c) Determinant is a number associated to a square matrix
 (d) All of the above
24. If α, β and γ are the roots of $x^3 + ax^2 + b = 0$, then the value of $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is
 (a) $-a^3$ (b) $a^3 - 3b$
 (c) a^3 (d) $a^2 - 3b$
25. If the axes are shifted to the point $(1, -2)$ without solution, then the equation $2x^2 + y^2 - 4x + 4y = 0$ becomes
 (a) $2X^2 + 3Y^2 = 6$
 (b) $2X^2 + Y^2 = 6$
 (c) $X^2 + 2Y^2 = 6$
 (d) None of the above
26. If $f(x) = \begin{cases} x^2, & x \leq 0 \\ 2 \sin x, & x > 0 \end{cases}$, then $x = 0$ is
 (a) point of minima
 (b) point of maxima
 (c) point of discontinuity
 (d) None of the above
27. In a group $(G, *)$, then equation $x * a = b$ has a
 (a) unique solution $b * a^{-1}$
 (b) unique solution $a^{-1} * b$
 (c) unique solution $a^{-1} * b^{-1}$
 (d) many solutions
28. A die is rolled twice and the sum of the numbers appearing on them is observed to be 7. What is the conditional probability that the number 2 has appeared at least once?
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{2}{3}$ (d) $\frac{2}{5}$
29. The locus of the mid-points of the focal chord of the parabola $y^2 = 4ax$ is
 (a) $y^2 = a(x - a)$ (b) $y^2 = 2a(x - a)$
 (c) $y^2 = 4a(x - a)$ (d) None of these
30. Find the value of $\sin 12^\circ \sin 48^\circ \sin 54^\circ$.
 (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) $\frac{1}{6}$ (d) $\frac{1}{8}$

31. In an equilateral triangle, the inradius, circumradius and one of the exradii are in the ratio
 (a) 2 : 3 : 5
 (b) 1 : 2 : 3
 (c) 1 : 3 : 7
 (d) 3 : 7 : 9
32. Let p and q be two statements. Then, $p \vee q$ is false, if
 (a) p is false and q is true
 (b) both p and q are false
 (c) both p and q are true
 (d) None of the above
33. In how many ways 6 letters be posted in 5 different letter boxes?
 (a) 5^6
 (b) 6^5
 (c) $5!$
 (d) $6!$
34. If A and B be two sets such that $A \times B$ consists of 6 elements. If three elements $A \times B$ are $(1, 4)$, $(2, 6)$ and $(3, 6)$, find $B \times A$.
 (a) $\{(1, 4), (1, 6), (2, 4), (2, 6), (3, 4), (3, 6)\}$
 (b) $\{(4, 1), (4, 2), (4, 3), (6, 1), (6, 2), (6, 3)\}$
 (c) $\{(4, 4), (6, 6)\}$
 (d) $\{(4, 1), (6, 2), (6, 3)\}$
35. Let $f : R \rightarrow R$ be defined as $f(x) = x^2 + 1$, find $f^{-1}(-5)$.
 (a) $\{\phi\}$
 (b) ϕ
 (c) $\{5\}$
 (d) $\{-5, 5\}$
36. If X is a poisson variate such that $P(X = 1) = P(X = 2)$, then $P(X = 4)$ is equal to
- (a) $\frac{1}{2e^2}$
 (b) $\frac{1}{3e^2}$
 (c) $\frac{2}{3e^2}$
 (d) $\frac{1}{e^2}$
37. The area enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is
 (a) 12 sq units
 (b) 13 sq units
 (c) $13\frac{1}{2}$ sq units
 (d) 14 sq units
38. The order and degree of the differential equation $\left(1 + 4\frac{dy}{dx}\right)^{2/3} = 4\frac{d^2y}{dx^2}$ are respectively
 (a) 1, $\frac{2}{3}$
 (b) 3, 2
 (c) 2, 3
 (d) 2, $\frac{2}{3}$
39. The solution of the differential equation $\frac{dy}{dx} = (4x + y + 1)^2$, is
 (a) $(4x + y + 1) = \tan(2x + C)$
 (b) $(4x + y + 1)^2 = 2 \tan(2x + C)$
 (c) $(4x + y + 1)^3 = 3 \tan(2x + C)$
 (d) $(4x + y + 1) = 2 \tan(2x + C)$
40. The system of equations $2x + y - 5 = 0$, $x - 2y + 1 = 0$, $2x - 14y - a = 0$, is consistent. Then, a is equal to
 (a) 1
 (b) 2
 (c) 5
 (d) None of these

Answer Key

1. a	2. c	3. c	4. a	5. c	6. d	7. a	8. d	9. b	10. d
11. d	12. a	13. a	14. b	15. b	16. a	17. b	18. b	19. b	20. a
21. a	22. a	23. c	24. c	25. b	26. a	27. a	28. b	29. b	30. d
31. b	32. b	33. a	34. b	35. b	36. c	37. d	38. c	39. d	40. d